Testing Evidence

MMM Social Workers & Associates — Booking System Application

Module	ITDMA3-34 — Project: Mobile Application and Web Services (Eduvos)
Group	Group 11
Group members	Faith Murau (EDUV4821116) | Khauhelo Moqechane (MD.2022.Q7M4P1) | Ignatius Mayibongwe Khumalo (EDUV4960805)
Purpose	Screenshot and recording evidence for every test ID recorded in TESTING.md
App version	1.0 (Android, Java; Firebase backend, region africa-south1)
Test period	From 15/10/2026 to 30/10/2026

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Results at a glance

Group	Total	PASS	PENDING
Automated unit tests	61	61	0
Hardening verification (PA1 to PF9)	37	37	0
Functional plan (F1 to F25)	25	25	0
Usability (U1 to U12)	12	12	0
Edge cases (E1 to E8)	8	8	0
Team: UX tests (UX-01 to UX-24)	24	0	24
Team: Navigation tests (NAV-01 to NAV-18)	18	0	18
Team: Database tests (DB-01 to DB-21)	21	0	21
Team: Booking logic tests (BK-01 to BK-22)	22	0	22
Team: Permission tests (PM-01 to PM-18)	18	0	18
Team: Integration tests (INT-01 to INT-12)	12	0	12

Update these numbers as tests are completed.

1. Automated tests

61 unit tests: ValidationUtilsTest (37) and BookingRulesTest (24). Run: right-click app/src/test/java > Run Tests.

UT1 — All unit tests pass

Test	Run every unit test in app/src/test/java
Expected result	61 tests, all green
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The Android Studio test results window showing 61 passed

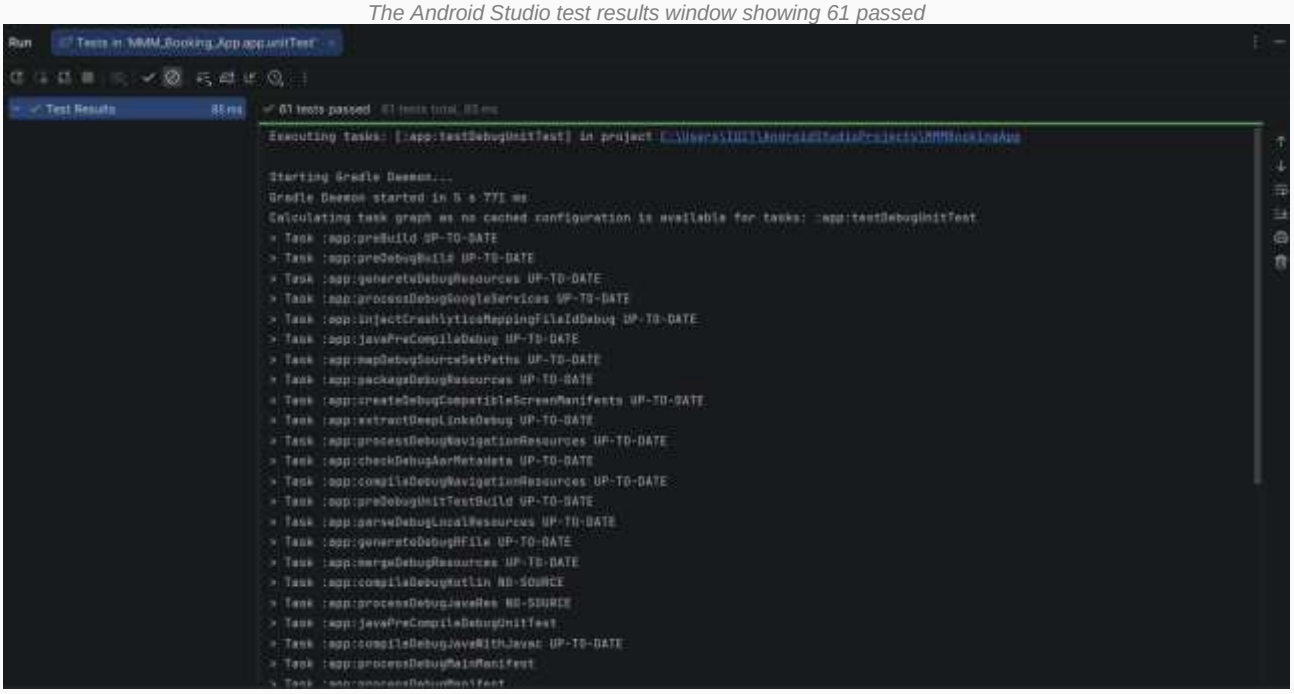
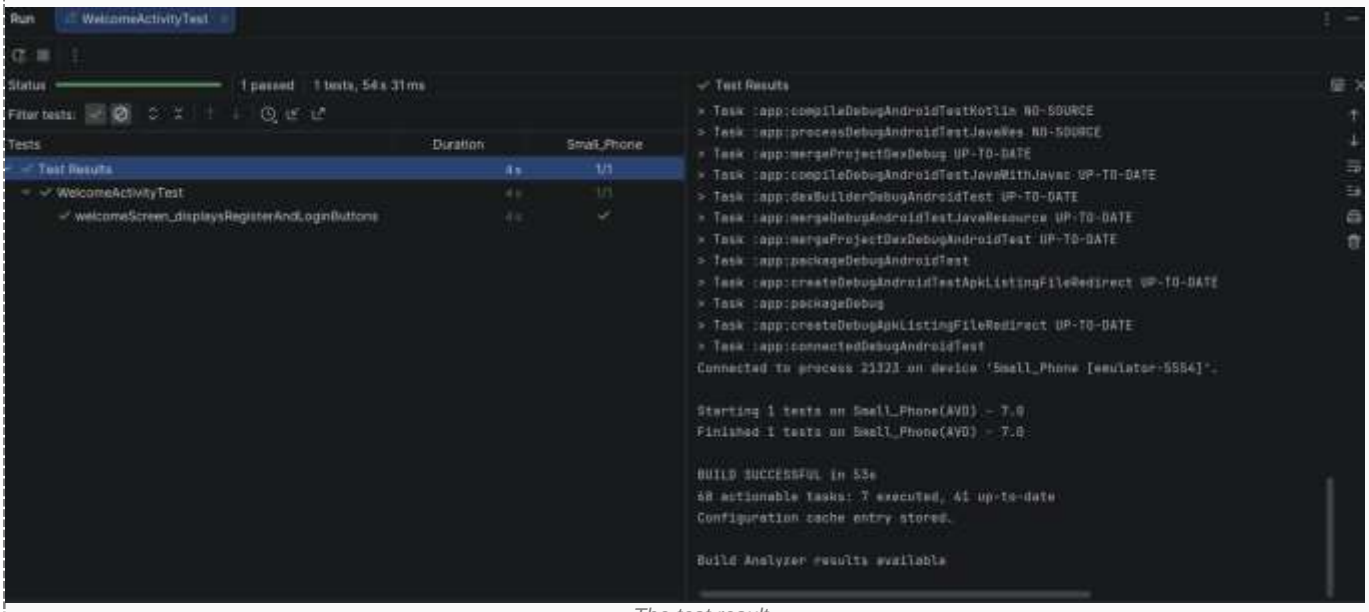


Figure UT1

UT2 — Espresso test on the Welcome screen

Test	Run WelcomeActivityTest on the connected device
Expected result	Register and Login buttons are visible and correctly labelled; test passes
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026



The test result

Figure UT2

2. Hardening verification tests (PA1 to PF9)

Security, data-integrity and robustness work from the final implementation audit. Each was run on the real phone and passed.

2.1 Phase A — Security rules

PA1 — Register a brand-new test account

Expected result	Account created normally; role defaults to client
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Registration success dialog

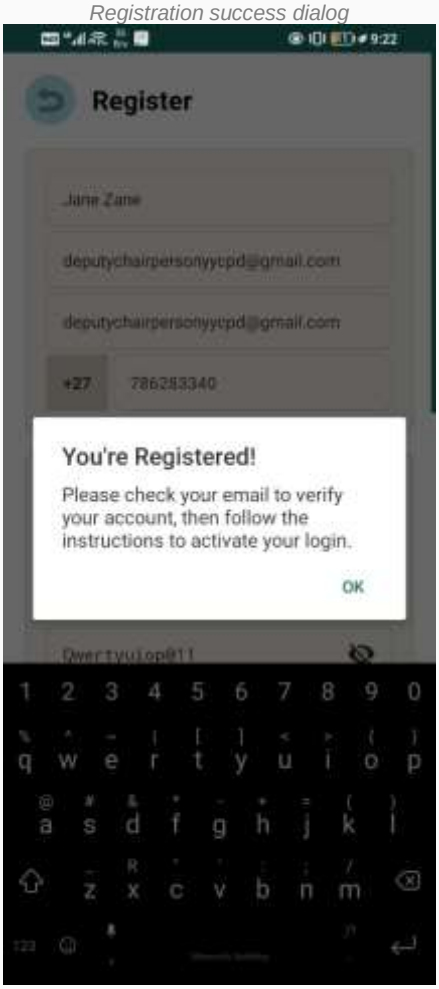


Figure PA1

PA2 — Confirm a client cannot set status/paid on a booking (Firebase Rules Playground)

Expected result	The write is denied by the rules
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Rules Playground screenshot showing the write denied



Figure PA2

PA3—Register, book and cancel as a client

Expected result	Whole flow still works under the new rules
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Booking confirmation, then the booking showing Cancelled in My Bookings

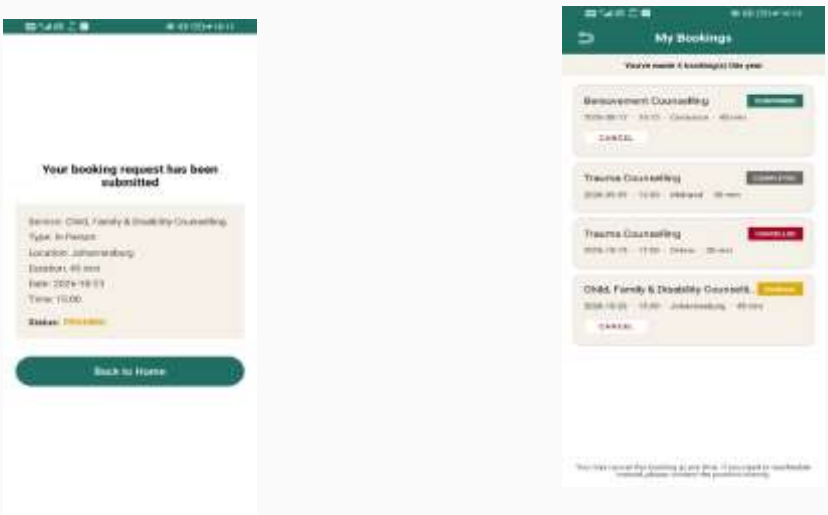


Figure PA3

2.2 Phase B — Booking CRUD (atomic cancel, true delete)

PB1—Client cancels a pending booking

Expected result	Shows Cancelled; the same time slot can be booked again
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Cancelled status in My Bookings, then the same slot booked again



Figure PB1

PB2—Admin declines a pending booking

Expected result	Confirmation dialog first; then Cancelled and slot freed
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Decline confirmation dialog, then the booking showing Cancelled

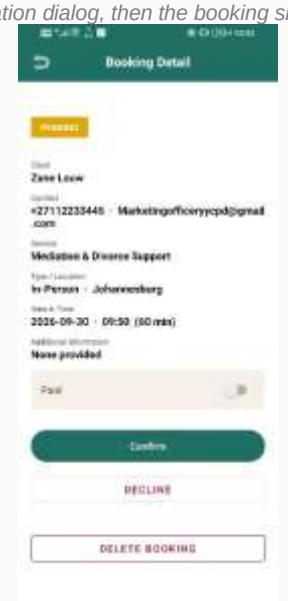


Figure PB2

PB3—Admin deletes a booking

Expected result	Confirmation first; screen closes; booking disappears from the dashboard; the bookings document is gone in the Firebase Console
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Delete dialog, dashboard without the booking, and the Firebase bookings collection without that document

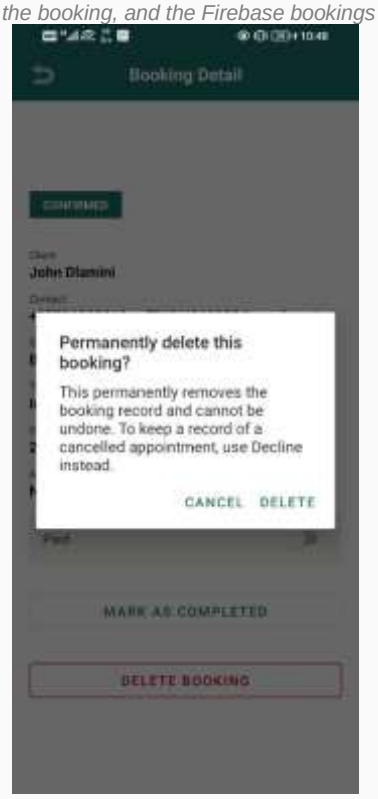


Figure PB3

PB4—Admin deletes a cancelled booking whose slot was since re-booked by another client

Expected result	The other client's slot document still exists
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Firebase bookedSlots showing the other client's slot still present after the delete

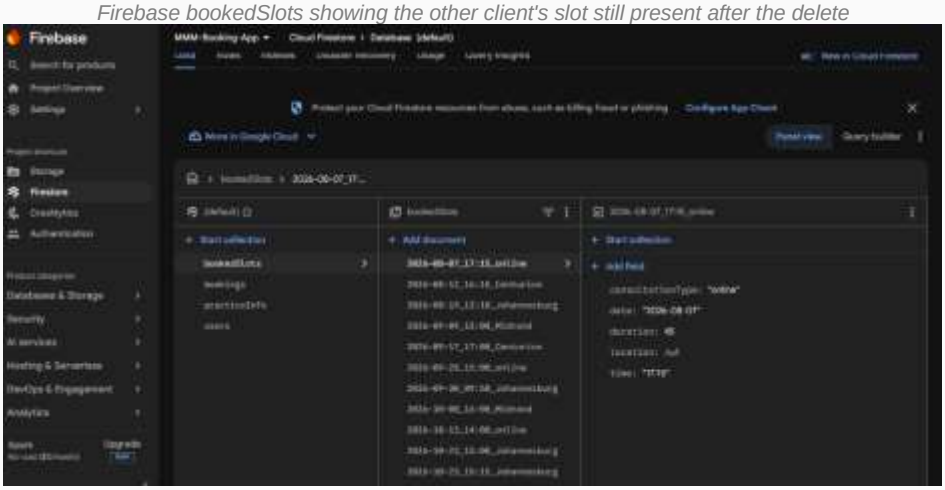


Figure PB4

PB5—Client has no delete action

Expected result	No client delete button; the server-side rule is the protection
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Client screens with no delete option



Figure PB5

2.3 Phase C — Booking rules

PC1 — Closing time (hours 08:00–18:00)

Expected result	60 min at 17:45 rejected; 17:00 accepted; 17:30 + 60 min rejected, switching to 30 min fits
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The rejection message for 17:45 + 60 min and the accepted 17:00 booking



Figure PC1

PC2 — Quarter-hour starts

Expected result	10:20 rejected with the quarter-hour message; 10:15 accepted
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The quarter-hour message for 10:20



Figure PC2

PC3 — Cross-client overlap

Expected result	Client A books 10:00 (60 min); Client B at 10:30 and at 10:00 are both rejected; 11:00 is accepted
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Client B's rejection message (both attempts) and the accepted 11:00 booking



Figure PC3

PC4 — Same-client overlap

Expected result	Client A booking 10:30 while holding 10:00–11:00 gets the 'you already have a booking that overlaps' message
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The own-overlap message



Figure PC4

PC5 — Slot freed on cancel

Expected result	After A cancels, B can book 10:00 straight away
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Client B's successful booking at 10:00 after A's cancellation

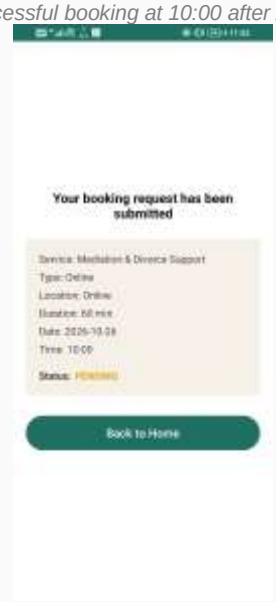


Figure PC5

PC6 — Operating days

Expected result	Untick Wednesday as admin; a client cannot pick a Wednesday; Contact shows 'Monday, Tuesday, Thursday - Friday'; restore afterwards
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Practice Information with Wednesday unticked, the closed-day message, and the Contact screen text.





Figure PC6

PC7 — Admin hours validation

Expected result	8:00, 17:00 and opening-after-closing each rejected; saving with no days ticked rejected
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Each validation message



Figure PC7

PC8 — Future-year holiday

Expected result	A date in next January is checked without error; 1 January is rejected
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The holiday-rejection message for 1 January



Figure PC8

2.4 Phase D — Booking Reminders switch

PD1 — Switch ON (default): book an appointment

Expected result	'Booking Submitted' notification appears
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The Booking Submitted notification

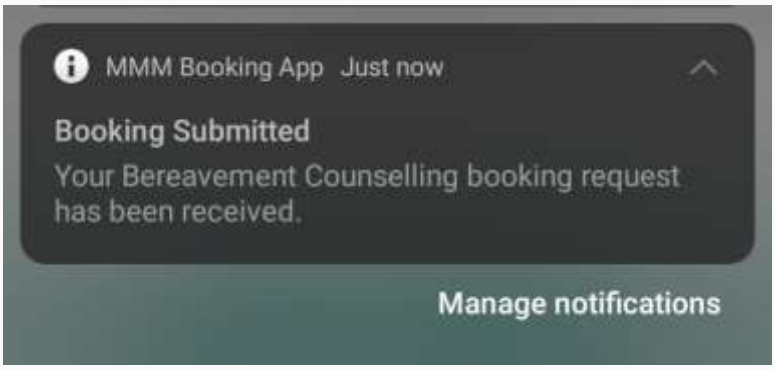


Figure PD1

PD2 — Switch OFF, then book

Expected result	'Disabled' toast; no notification; booking still succeeds
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The disabled toast and the absence of a notification after booking

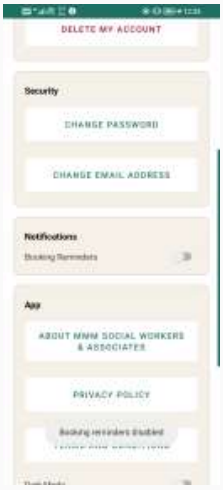


Figure PD2

PD3 — Re-open Settings several times

Expected result	No toast on opening — a toast only appears when the switch is tapped
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Settings opened with no toast)

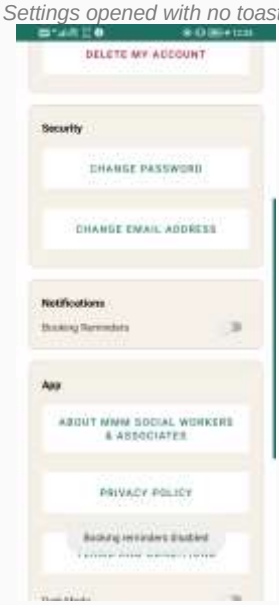


Figure PD3

PD4 — Scheduled reminder cancelled when OFF, restored when ON (adb shell dumpsys alarm)

Expected result	Alarm present when ON, gone when OFF, back when ON again
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The dumpsys output for ON, OFF and ON again

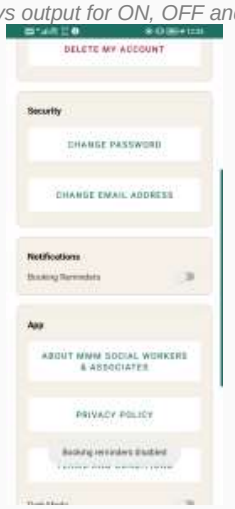


Figure PD4

PD5 — Persistence

Expected result	Still OFF after fully closing the app; notificationsEnabled is false in the Console
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Settings showing OFF after reopening, and the Console field



Figure PD5

PD6 — Account isolation

Expected result	A second account on the same phone shows the switch ON
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The second account's Settings with the switch ON



Figure PD6

PD7 — Dark Mode toggle

Expected result	Still works as before
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Dark Mode on and off



Figure PD7

2.5 Phase E — Robustness

PE1 — Registration rollback (profile creation temporarily blocked in the rules)

Expected result	'Registration failed' toast; the new email is not left in Authentication > Users
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The failure toast and the Authentication Users list without the new email



Figure PE1

PE2 — Delete a booking's status field in the Console

Expected result	Admin Dashboard does not crash; booking shown/counted as Pending; Booking Detail opens
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The dashboard with the booking counted as Pending, and its Booking Detail



Figure PE2

PE3 — Reopen the app in Airplane Mode (client and admin)

Expected result	Lands on Home / Admin Dashboard, not Welcome
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Home and Admin Dashboard opened in Airplane Mode



Figure PE3

PE4 — Regression: login routing and back button

Expected result	Client to Home, admin to Dashboard with the 'Logged in successfully' toast; double-back-to-exit works
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The success toast on each home screen, and the back-press toast



Figure PE4

PE5 — Logcat check

Expected result	No email address in any LoginActivity/RegisterActivity log line
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Logcat filtered to LoginActivity and RegisterActivity



Figure PE5

2.6 Phase F — Clean-up and validation

PF1 — Clean build after deleting the default test stubs

Expected result	Compiles with no errors
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Successful build output

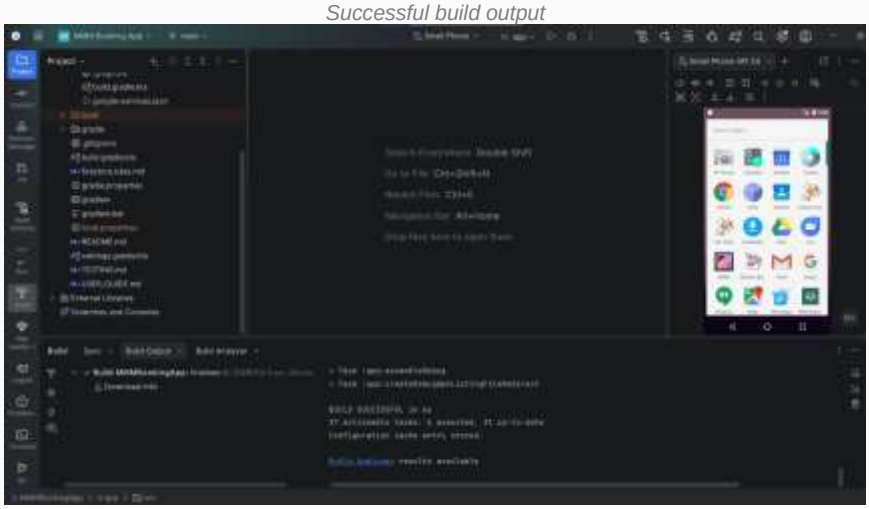


Figure PF1

PF2 — Run all unit tests

Expected result	61 tests, all green
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The Android Studio test results window showing 61 passed

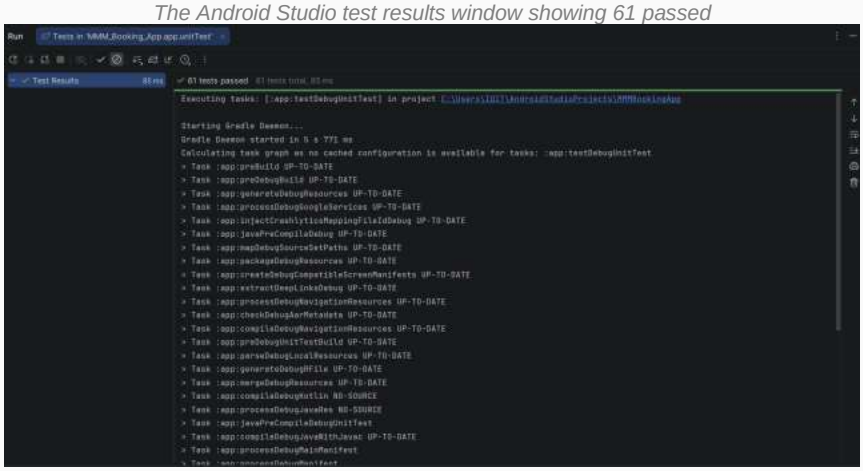


Figure PF2

PF3 — Register with email 'thandi@example'

Expected result	Email error shown; thandi@example.com accepted
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The email error for thandi@example

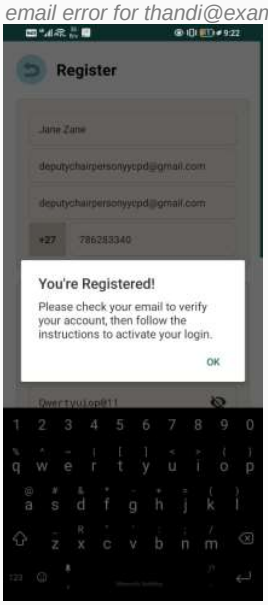


Figure PF3

PF4 — Register with a weak password

Expected result	Rules line turns red and bold, no toast; normal when typing, green when valid
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The red, bold password rules line after submit



Figure PF4

PF5 — Log In with empty fields

Expected result	'This field is required' on each empty field
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

The required-field messages on Login



Figure PF5

PF6 — Normal login

Expected result	Client to Home, admin to Dashboard with the success toast
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Home and Admin Dashboard after login





Figure PF6

PF7 — Booking Detail as admin

Expected result	Badge colour and Confirm/Decline/Complete/Delete behave as before; confirmation dialogs appear
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Booking Detail as admin



Figure PF7

PF8 — Comment-only files (Home, Change Email, Admin Analytics)

Expected result	Behave exactly as before
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Home, Change Email and Admin Analytics screens



Figure PF8

PF9 — git status after untracking .idea

Expected result	.idea no longer tracked
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

Terminal output of git status



Figure PF9

3. Functional test plan (F1 to F25)

F1 — Registration

Steps	Fill all fields correctly, tick consent, submit
Expected result	Account created, verification email sent
Result	**PASS (PA1)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F1

F2 — Registration blocked without consent

Steps	Fill all fields, leave consent unticked, submit
Expected result	Red error shown, no account created
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F2

F3 — Email verification gate

Steps	Log in before verifying email
Expected result	Blocked, resend option shown
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

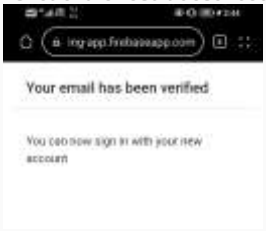


Figure F3

F4 — Login (verified account)

Steps	Log in with correct credentials
Expected result	Routed to Home (client) or Admin Dashboard (admin), with a brief 'Logged in successfully' toast
Result	**PASS (PE4, PF6)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

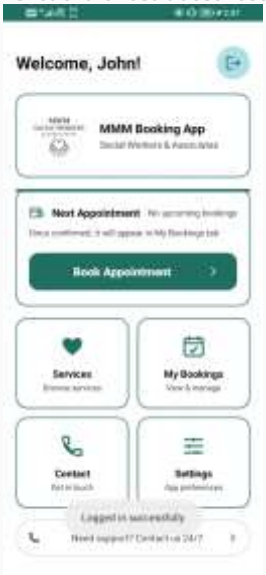


Figure F4

F5 — Password reset

Steps	Request reset for a registered email
Expected result	Reset email received
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F5

F6 — Booking submission

Steps	Complete and submit a valid booking
Expected result	Confirmation screen shown, booking appears in My Bookings, immediate notification received
Result	**PASS (PD1)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

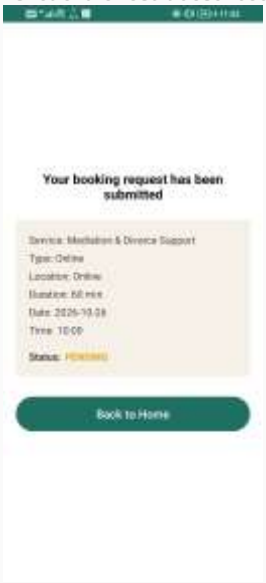


Figure F6

F7 — Double-booking prevention

Steps	Two accounts attempt overlapping times
Expected result	Second attempt rejected
Result	**PASS (PC3)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F7

F8 — Public holiday block

Steps	Attempt a known SA public holiday date
Expected result	Booking date rejected with a clear message
Result	**PASS (PC8)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F8

F9 — Cancellation

Steps	Cancel a pending/confirmed booking
Expected result	Confirmation dialog shown, status becomes Cancelled, slot freed
Result	**PASS (PB1)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F9

F10 — Admin confirm/decline

Steps	Admin confirms a pending booking
Expected result	Status updates automatically on the client's My Bookings screen in real time
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

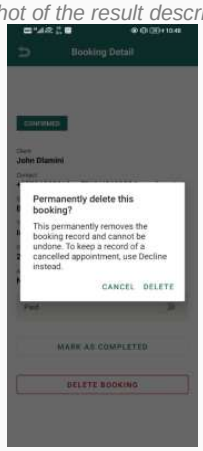


Figure F10

F11 — Admin search

Steps	Type a service name into the search box
Expected result	List filters to matching bookings only
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F11

F12 — Admin search + filters combined

Steps	Apply a status filter, then search
Expected result	Only bookings matching both are shown
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F12

F13 — CSV export

Steps	Tap Export CSV on Admin Dashboard
Expected result	Share sheet opens with a valid CSV file
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F13

F14 — Practice Information edit

Steps	Admin changes open days/phone, saves
Expected result	Client's Contact and booking screens reflect the change
Result	**PASS (PC6)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above





Figure F14

F15 — Admin Analytics

Steps	Open View Detailed Analytics
Expected result	Counts, % change and bar charts display for both Bookings Created and New Registrations
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F15

F16 — Settings — single logout

Steps	Open Settings, scroll to Support
Expected result	Exactly one Logout button present
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F16

F17 — Dark Mode toggle confirmation

Steps	Toggle Dark Mode on Settings
Expected result	Toast confirms enabled/disabled; theme changes across the app
Result	**PASS (PD7)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F17

F18 — Booking Reminders toggle

Steps	Toggle Booking Reminders on Settings
Expected result	Toast confirms enabled/disabled; notifications obey the switch
Result	**PASS (PD2, PD3)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

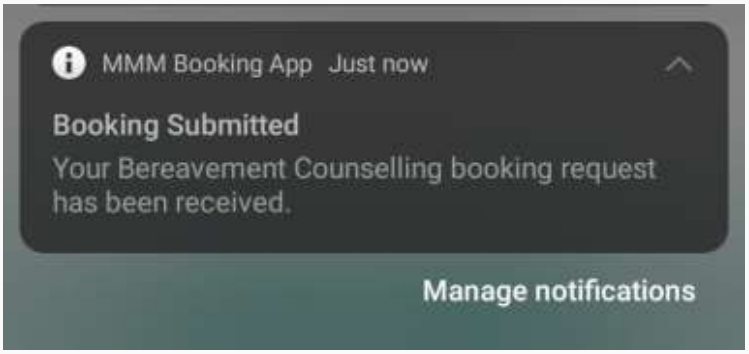


Figure F18

F19 — Logout confirmation dialog

Steps	Tap Logout from Home, Settings, and Admin Dashboard
Expected result	Cancel/Log Out dialog each time; Cancel does nothing; Log Out returns to Welcome
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F19

F20 — Delete Account explainer

Steps	Tap Delete My Account on Settings
Expected result	Explainer first; Continue opens the pre-addressed email; Cancel dismisses
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above

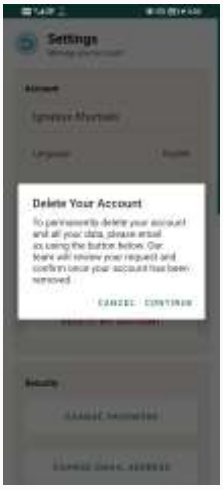


Figure F20

F21 — Back button on Home/Admin Dashboard

Steps	Press back once, then again within 2 seconds
Expected result	First press shows 'Press back again to exit'; second exits; user stays logged in
Result	**PASS (PE4)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F21

F22 — Session persistence

Steps	Log in, fully close the app, reopen
Expected result	Opens directly to Home/Admin Dashboard without logging in again
Result	**PASS (PE3)**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F22

F23 — Real-time admin list update

Steps	With Admin Dashboard open, a client submits a booking from another device
Expected result	New booking and updated counts appear automatically
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F23

F24 — Confirm-email mismatch on Register

Steps	Type different addresses in Email and Confirm Email, submit
Expected result	'Email addresses do not match' shown; no account created
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F24

F25 — Change Email

Steps	Settings > Change Email with current password and a new address
Expected result	'Confirmation Sent' dialog; link arrives at the new address; login email unchanged until clicked
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

A screenshot of the result described above



Figure F25

4. Usability tests (U1 to U12)

U1 — Dropdown fields

Expected result	Rounded border + visible arrow on all Spinners
Result	**PASS**
Device used	Phone and Table Date: 15/10/2026

INSERT SCREENSHOT FOR U1 HERE
A screenshot showing the point being checked (phone and/or tablet)

Figure U1

U2 — Password fields

Expected result	Eye icon toggles visibility correctly on Login and Register
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR U2 HERE
A screenshot showing the point being checked (phone and/or tablet)

Figure U2

U3 — Dark Mode

Expected result	All screens switch palette, remain readable; status badges show bold white text with good contrast
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR U3 HERE

A screenshot showing the point being checked (phone and/or tablet)

Figure U3

U4 — Scroll indicators

Expected result	Visible on screens with content exceeding one view
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR U4 HERE

A screenshot showing the point being checked (phone and/or tablet)

Figure U4

U5 — Touch target size

Expected result	Admin Dashboard's Logout and Settings icons comfortably tappable
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR U5 HERE

A screenshot showing the point being checked (phone and/or tablet)

Figure U5

U6 — Register checkbox text

Expected result	Both consent sentences fully visible; '(optional)' has clear spacing below it
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR U6 HERE

A screenshot showing the point being checked (phone and/or tablet)

Figure U6

U7 — Settings long labels

Expected result	'About', 'FAQ' etc. wrap to two lines cleanly, never cut off
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR U7 HERE

A screenshot showing the point being checked (phone and/or tablet)

Figure U7

U8 — Logout dialog wording

Expected result	States that logging out requires logging in again; button order Cancel (left), Log Out (right)
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR U8 HERE

A screenshot showing the point being checked (phone and/or tablet)

Figure U8

U9 — Tablet layout — centred content

Expected result	On a tablet, content is centred with generous margins, not stretched edge-to-edge
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR U9 HERE

A screenshot showing the point being checked (phone and/or tablet)

Figure U9

U10 — Tablet layout — no dead space

Expected result	Fixed-content screens balance vertically on a tall tablet screen
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR U10 HERE

A screenshot showing the point being checked (phone and/or tablet)

Figure U10

U11 — Tablet font scaling

Expected result	Text is visibly larger on tablet than on phone, without layout breaking
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR U11 HERE

A screenshot showing the point being checked (phone and/or tablet)

Figure U11

U12 — Landscape orientation

Expected result	Home, Book Appointment, Register remain usable and scrollable in landscape on phone and tablet
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR U12 HERE

A screenshot showing the point being checked (phone and/or tablet)

Figure U12

5. Edge-case tests (E1 to E8)

E1 — Submit booking form with fields missing

Expected result	Specific message per missing field
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR E1 HERE
A screenshot of the message or screen shown

Figure E1

E2 — Attempt booking while offline

Expected result	Clear offline message, no crash
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR E2 HERE
A screenshot of the message or screen shown

Figure E2

E3 — Attempt login while offline

Expected result	Clear offline message, no crash
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR E3 HERE

A screenshot of the message or screen shown

Figure E3

E4 — Open My Bookings offline, never loaded before

Expected result	Empty state shown, no crash
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR E4 HERE

A screenshot of the message or screen shown

Figure E4

E5 — Holiday API unreachable

Expected result	Booking still works on open-day and hours rules; toast notes holidays couldn't be checked
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR E5 HERE

A screenshot of the message or screen shown

Figure E5

E6 — Search with no matching results

Expected result	Empty state shown, no crash
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR E6 HERE

A screenshot of the message or screen shown

Figure E6

E7 — Export CSV with zero bookings in the filtered list

Expected result	Export still runs with a header row only
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR E7 HERE

A screenshot of the message or screen shown

Figure E7

E8 — Back button on a non-root screen (e.g. Book Appointment)

Expected result	Normal back navigation — the 'press again to exit' behaviour applies only to Home/Admin Dashboard
Result	**PASS** (screenshot still to be inserted below)
Device used	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR E8 HERE

A screenshot of the message or screen shown

Figure E8

6. Team test plans

Test plans assigned to each group member. Every test is PENDING until it is run; mark PASS or FAIL and insert the evidence under it. Each member's evidence pack is described at the start of their section.

6.1 UX/UI testing — Faith Murau

Goal
Prove the app looks consistent, is easy to understand, and gives clear feedback. Test on at least two screen sizes and in both light and dark mode (Settings has a dark mode switch).

Evidence pack
Side-by-side screenshots of every screen in light and dark mode, a table of every message with the clear/vague/technical rating, and a short list of recommended UI improvements.

Visual consistency

UX-01

Owner	Faith Murau
Steps / what to check	Colour palette across all 22 screens
Expected result	Same primary colour, background and text colours everywhere; status colours consistent (pending, confirmed, cancelled, completed each has one colour).
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-01 HERE
Light-mode and dark-mode screenshots (and a second screen size where relevant)

Figure UX-01

UX-02

Owner	Faith Murau
Steps / what to check	Fonts and sizes
Expected result	Titles, body text and button labels use consistent sizes; no text smaller than readable (about 12sp).
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-02 HERE

Light-mode and dark-mode screenshots (and a second screen size where relevant)

Figure UX-02

UX-03

Owner	Faith Murau
Steps / what to check	Buttons
Expected result	Primary buttons look the same on every screen; disabled buttons look disabled (for example Submit while loading).
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-03 HERE

Light-mode and dark-mode screenshots (and a second screen size where relevant)

Figure UX-03

UX-04

Owner	Faith Murau
Steps / what to check	Spacing and alignment
Expected result	Equal margins left and right, no overlapping or clipped text, nothing cut off at the bottom.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-04 HERE

Light-mode and dark-mode screenshots (and a second screen size where relevant)

Figure UX-04

UX-05

Owner	Faith Murau
Steps / what to check	Dark mode
Expected result	Switch on in Settings. Every screen stays readable: no dark text on dark background, no white blocks. Setting survives closing and reopening the app.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-05 HERE

Light-mode and dark-mode screenshots (and a second screen size where relevant)

Figure UX-05

UX-06

Owner	Faith Murau
Steps / what to check	Screen sizes and rotation
Expected result	Small phone and tablet both look correct. Rotate on Register, Book Appointment and Admin Dashboard: fields stay filled and nothing crashes.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-06 HERE

Light-mode and dark-mode screenshots (and a second screen size where relevant)

Figure UX-06

UX-07

Owner	Faith Murau
Steps / what to check	Touch targets
Expected result	Buttons and icons are easy to tap with a thumb (about 48dp).
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-07 HERE

Light-mode and dark-mode screenshots (and a second screen size where relevant)

Figure UX-07

UX-08

Owner	Faith Murau
Steps / what to check	Branding
Expected result	App name, logo and colours match on Welcome, Login, Home and About.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-08 HERE

Light-mode and dark-mode screenshots (and a second screen size where relevant)

Figure UX-08

Clarity and ease of use

UX-09

Owner	Faith Murau
Steps / what to check	First-time user test: ask someone who has never seen the app to register, book and cancel without help. Time them and note where they hesitate.
Expected result	The task is completed without help; hesitation points are recorded.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-09 HERE

Light-mode and dark-mode screenshots (and a second screen size where relevant)

Figure UX-09

UX-10

Owner	Faith Murau
Steps / what to check	Labels and hints
Expected result	Every input has a clear label or hint (phone shows that the +27 prefix is fixed, password requirements are visible).
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-10 HERE

Light-mode and dark-mode screenshots (and a second screen size where relevant)

Figure UX-10

UX-11

Owner	Faith Murau
Steps / what to check	Booking screen flow
Expected result	Service, type, location (only when In-Person), duration, date, time, notes: order makes sense, Online shows a note instead of a location.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-11 HERE

Light-mode and dark-mode screenshots (and a second screen size where relevant)

Figure UX-11

UX-12

Owner	Faith Murau
Steps / what to check	Empty states: a new client opens My Bookings; the Admin Dashboard with no bookings
Expected result	A friendly empty message appears on each.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-12 HERE

Light-mode and dark-mode screenshots (and a second screen size where relevant)

Figure UX-12

UX-13

Owner	Faith Murau
Steps / what to check	Loading feedback
Expected result	Progress indicator shows while booking, logging in and loading lists; the user is never left looking at a frozen screen.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-13 HERE

Light-mode and dark-mode screenshots (and a second screen size where relevant)

Figure UX-13

UX-14

Owner	Faith Murau
Steps / what to check	Offline banner: turn off data, then turn it back on
Expected result	A banner appears on supported screens; booking is blocked with a clear message. When data is back on, the banner disappears.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-14 HERE

Light-mode and dark-mode screenshots (and a second screen size where relevant)

Figure UX-14

Error and confirmation messages (record the exact message and rate it: clear, vague or technical)

UX-15

Owner	Faith Murau
Steps / what to check	Register with a one-word name
Expected result	Inline error that two names are needed.
Message shown / rating	Exact message: _____ ☐ clear ☐ vague ☐ technical
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-15 HERE

Screenshot of the message, plus the exact wording

Figure UX-15

UX-16

Owner	Faith Murau
Steps / what to check	Register with 8 digits in phone
Expected result	Inline phone error (9 digits required after +27).
Message shown / rating	Exact message: _____ ☐ clear ☐ vague ☐ technical
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-16 HERE
Screenshot of the message, plus the exact wording

Figure UX-16

UX-17

Owner	Faith Murau
Steps / what to check	Register with mismatched email or password confirmation
Expected result	Inline mismatch errors.
Message shown / rating	Exact message: _____ ☐ clear ☐ vague ☐ technical
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-17 HERE
Screenshot of the message, plus the exact wording

Figure UX-17

UX-18

Owner	Faith Murau
Steps / what to check	Register without ticking consent
Expected result	Red consent error, no account created.
Message shown / rating	Exact message: _____ ☐ clear ☐ vague ☐ technical
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-18 HERE
Screenshot of the message, plus the exact wording

Figure UX-18

UX-19

Owner	Faith Murau
Steps / what to check	Register with a weak password
Expected result	The password rules line turns red and bold next to the field (no toast), consistent with the other fields.
Message shown / rating	Exact message: _____ ☐ clear ☐ vague ☐ technical
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-19 HERE
Screenshot of the message, plus the exact wording

Figure UX-19

UX-20

Owner	Faith Murau
Steps / what to check	Login with empty fields
Expected result	'This field is required' appears on each empty field.
Message shown / rating	Exact message: _____ ☐ clear ☐ vague ☐ technical
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-20 HERE
Screenshot of the message, plus the exact wording

Figure UX-20

UX-21

Owner	Faith Murau
Steps / what to check	Login with wrong password
Expected result	Clear failure message that does not reveal which part was wrong.
Message shown / rating	Exact message: _____ ☐ clear ☐ vague ☐ technical
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-21 HERE
Screenshot of the message, plus the exact wording

Figure UX-21

UX-22

Owner	Faith Murau
Steps / what to check	Booking on a closed day, a public holiday, and outside hours
Expected result	Specific messages naming the open days, the holiday, or the opening and closing times.
Message shown / rating	Exact message: _____ ☐ clear ☐ vague ☐ technical
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-22 HERE

Screenshot of the message, plus the exact wording

Figure UX-22

UX-23

Owner	Faith Murau
Steps / what to check	Booking on a taken time
Expected result	Clear message that the time is already reserved.
Message shown / rating	Exact message: _____ ☐ clear ☐ vague ☐ technical
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-23 HERE

Screenshot of the message, plus the exact wording

Figure UX-23

UX-24

Owner	Faith Murau
Steps / what to check	Successful booking, cancel, decline, delete, paid toggle
Expected result	A confirmation message appears for each, and destructive actions (cancel, decline, delete) show a confirmation dialog first.
Message shown / rating	Exact message: _____ ☐ clear ☐ vague ☐ technical
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR UX-24 HERE

Screenshot of the message, plus the exact wording

Figure UX-24

6.2 Screen and navigation testing — Khauhelo Moqechane

Goal

Every screen opens, every button does something, and the back stack behaves.

Evidence pack

The completed inventory table, a navigation flow diagram (draw the screen graph that you actually observed), a list of broken or missing items, and a short screen recording of the full client journey and the admin journey.

Screen inventory (open each screen once and tick it off)

Screen	Reached from	Notes	Opened
Splash / router	App launch	Routes to Welcome, client Home or admin Dashboard depending on session	☐
Welcome	Splash (logged out)		☐
Login	Welcome		☐
Register	Welcome		☐
Forgot Password	Login		☐
Home	Login (client)		☐
Services	Home		☐
Book Appointment	Home or Services		☐

Screen	Reached from	Notes	Opened
Booking Confirmation	After a successful booking		☐
My Bookings	Home		☐
Contact	Home		☐
Settings	Home or menu		☐
Change Password / Change Email	Settings		☐
FAQ / About / Privacy / Terms	Settings or Register (terms)		☐
Admin Dashboard	Login (admin)		☐
Booking Detail	Dashboard list item		☐
Practice Information	Dashboard		☐
Admin Analytics	Dashboard		☐

Broken buttons and missing screens

Tap every button, icon, list item and link on every screen, including the back arrows. Log any that do nothing, crash, or go to the wrong place. Then compare against the original wireframes from the earlier deliverable and list any screen in the wireframes that is missing from the app. Design note: Delete Account in Settings does not delete data in the app; it opens an email to the practice. Record whether the label makes that clear.

Navigation tests

NAV-01

Owner	Khauhelo Moqechane
Steps / what to check	Fresh install, open app
Expected result	Welcome screen with working Login and Register buttons.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-01 HERE

Screenshots or a short screen recording of the steps and the result

Figure NAV-01

NAV-02

Owner	Khauhelo Moqechane
Steps / what to check	Login with a verified client
Expected result	Lands on Home. Pressing back does not return to Login.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-02 HERE

Screenshots or a short screen recording of the steps and the result

Figure NAV-02

NAV-03

Owner	Khauhelo Moqechane
Steps / what to check	Login with admin
Expected result	Lands on Admin Dashboard, not client Home.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-03 HERE

Screenshots or a short screen recording of the steps and the result

Figure NAV-03

NAV-04

Owner	Khauhelo Moqechane
Steps / what to check	Close app (swipe away) while logged in, reopen
Expected result	Goes straight to the correct home screen for the role, no login required.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-04 HERE

Screenshots or a short screen recording of the steps and the result

Figure NAV-04

NAV-05

Owner	Khauhelo Moqechane
Steps / what to check	Log out, press back
Expected result	Cannot return to a logged-in screen.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-05 HERE

Screenshots or a short screen recording of the steps and the result

Figure NAV-05

NAV-06

Owner	Khauhelo Moqechane
Steps / what to check	Welcome to Register to Login using on-screen links, then back
Expected result	Back moves one step at a time with no crash or blank screen.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-06 HERE

Screenshots or a short screen recording of the steps and the result

Figure NAV-06

NAV-07

Owner	Khauhelo Moqechane
Steps / what to check	Register successfully
Expected result	Verification message shown, user returned to a sensible screen (Login), not left in a half-logged-in state.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-07 HERE

Screenshots or a short screen recording of the steps and the result

Figure NAV-07

NAV-08

Owner	Khauhelo Moqechane
Steps / what to check	Login, then Forgot Password, then back
Expected result	Returns to Login with fields intact.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-08 HERE

Screenshots or a short screen recording of the steps and the result

Figure NAV-08

NAV-09

Owner	Khauhelo Moqechane
Steps / what to check	Home, then Book Appointment, then submit
Expected result	Booking Confirmation opens; back from Confirmation goes Home or My Bookings, not back into the filled booking form (the form is closed on submit).
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-09 HERE

Screenshots or a short screen recording of the steps and the result

Figure NAV-09

NAV-10

Owner	Khauhelo Moqechane
Steps / what to check	Home, then My Bookings, cancel a booking, then back
Expected result	List already updated (live listener), back goes Home.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-10 HERE

Screenshots or a short screen recording of the steps and the result

Figure NAV-10

NAV-11

Owner	Khauhelo Moqechane
Steps / what to check	Settings, then Change Password, back; Change Email, back
Expected result	Returns to Settings each time.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-11 HERE

Screenshots or a short screen recording of the steps and the result

Figure NAV-11

NAV-12

Owner	Khauhelo Moqechane
Steps / what to check	Terms/Privacy opened from Register, then back
Expected result	Registration form keeps its typed values.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-12 HERE

Screenshots or a short screen recording of the steps and the result

Figure NAV-12

NAV-13

Owner	Khauhelo Moqechane
Steps / what to check	Admin Dashboard, then Booking Detail, confirm, back
Expected result	Dashboard shows the updated status.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-13 HERE

Screenshots or a short screen recording of the steps and the result

Figure NAV-13

NAV-14

Owner	Khauhelo Moqechane
Steps / what to check	Admin Dashboard: press back once, then again quickly
Expected result	First press shows an exit hint, second exits (double-press behaviour).
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-14 HERE
Screenshots or a short screen recording of the steps and the result

Figure NAV-14

NAV-15

Owner	Khauhelo Moqechane
Steps / what to check	Rotate device on every major screen
Expected result	No crash; data preserved or sensibly reloaded.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-15 HERE
Screenshots or a short screen recording of the steps and the result

Figure NAV-15

NAV-16

Owner	Khauhelo Moqechane
Steps / what to check	Rapidly tap a button twice (Login, Submit Booking)
Expected result	Only one action happens (button disabled while loading); no duplicate booking.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-16 HERE

Screenshots or a short screen recording of the steps and the result

Figure NAV-16

NAV-17

Owner	Khauhelo Moqechane
Steps / what to check	Open app from a notification (reminder or booking submitted)
Expected result	Opens the app without a crash.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-17 HERE

Screenshots or a short screen recording of the steps and the result

Figure NAV-17

NAV-18

Owner	Khauhelo Moqechane
Steps / what to check	Turn off data, then tap navigation items
Expected result	Screens open; the offline banner shows; no crash.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR NAV-18 HERE

Screenshots or a short screen recording of the steps and the result

Figure NAV-18

6.3 Database, authentication and booking logic — Ignatius Mayibongwe Khumalo

Goal

Prove the backend rules work, data is stored correctly and access control cannot be bypassed. Keep the Firebase Console open beside the phone and check the actual documents after every action.

Evidence pack

Firestore Console screenshots before and after each action, the Rules Playground result for every PM test, and a table of the stored fields against the data design from Deliverable 1.

Validation rules to test against

Password: at least 8 characters with upper case, lower case, a digit and a special character. Phone: +27 fixed prefix and exactly 9 digits. Name: at least two words of at least two letters.

Authentication (DB-01 to DB-13)

DB-01

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Register Client A with valid details
Expected result	Firebase Authentication shows the new user; users/{uid} exists with name, email, phone and role: "client"; verification email arrives.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-01 HERE

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

Figure DB-01

DB-02

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Login before verifying email
Expected result	Blocked, with an option to resend the verification email.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-02 HERE

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

Figure DB-02

DB-03

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Verify via email link, log in
Expected result	Succeeds and routes to Home.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-03 HERE

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

Figure DB-03

DB-04

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Wrong password; unregistered email
Expected result	Login fails with a message; no session created.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-04 HERE

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

Figure DB-04

DB-05

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Register with an already used email
Expected result	Clear failure, no duplicate user document.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-05 HERE

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

Figure DB-05

DB-06

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Boundary data: password with 7 chars, no uppercase, no digit, no special; phone 8 and 10 digits; name "A B"
Expected result	Each is rejected.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-06 HERE

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

Figure DB-06

DB-07

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Email without a proper domain, for example abc@ or a@b
Expected result	Rejected with the email error (the email format is checked, not just the @ sign).
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-07 HERE

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

Figure DB-07

DB-08

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Forgot Password with a registered email
Expected result	Reset email received and the new password works.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-08 HERE

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

Figure DB-08

DB-09

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Change Password with correct old password; with wrong old password
Expected result	Succeeds / rejected. Login with the new password works.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-09 HERE

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

Figure DB-09

DB-10

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Change Email
Expected result	Verification flow works and the users document email stays consistent.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-10 HERE

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

Figure DB-10

DB-11

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Logout, then relaunch
Expected result	Back at Welcome, no session.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-11 HERE

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

Figure DB-11

DB-12

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Register while offline
Expected result	Fails with a message and no orphaned account is left in Firebase Authentication.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-12 HERE

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

Figure DB-12

DB-13

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Login as the admin account
Expected result	Routed to the admin dashboard; client accounts never reach it.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-13 HERE

App screenshot plus the matching Firebase Console view (Authentication / Firestore)

Figure DB-13

Data saved and retrieved (DB-14 to DB-21)

DB-14

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A books In-Person, 60 min, with notes
Expected result	bookings/{id} has clientId = A's uid, service, consultationType in-person, location, duration 60, date yyyy-MM-dd, time HH:mm, additionalInfo, status pending, paid false, createdAt.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-14 HERE

App screenshot plus the Firestore document showing the stored fields

Figure DB-14

DB-15

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Same booking in My Bookings
Expected result	All details match what was entered.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-15 HERE

App screenshot plus the Firestore document showing the stored fields

Figure DB-15

DB-16

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Book Online
Expected result	location is null and consultationType is online.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-16 HERE

App screenshot plus the Firestore document showing the stored fields

Figure DB-16

DB-17

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Admin opens Booking Detail
Expected result	Shows client name, phone and email read from users, plus all booking fields.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-17 HERE

App screenshot plus the Firestore document showing the stored fields

Figure DB-17

DB-18

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Admin confirms, marks paid, marks completed
Expected result	Firestore status and paid change each time and Client A sees the new status live without refreshing.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-18 HERE

App screenshot plus the Firestore document showing the stored fields

Figure DB-18

DB-19

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Edit Practice Information (days, hours)
Expected result	practiceInfo document updates and the Book Appointment screen immediately uses the new hours.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-19 HERE

App screenshot plus the Firestore document showing the stored fields

Figure DB-19

DB-20

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Practice hours validation: end before start; malformed times; no open days
Expected result	Each rejected with a message; nothing saved.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-20 HERE

App screenshot plus the Firestore document showing the stored fields

Figure DB-20

DB-21

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Settings: toggle reminders and dark mode
Expected result	Preference saved on the account/device and still applied after restart.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR DB-21 HERE

App screenshot plus the Firestore document showing the stored fields

Figure DB-21

6.4 Booking logic — Ignatius Mayibongwe Khumalo

How slots work

A booking reserves one bookedSlots document per 15-minute block, named date_time_location (location is "online" for online bookings). A 60-minute booking at 10:00 reserves 10:00, 10:15, 10:30 and 10:45.

BK-01

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Book 10:00 for 60 min
Expected result	Four slot documents appear with Client A's uid as clientId.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-01 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-01

BK-02

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client B books the same location, same date, 10:00
Expected result	Rejected: time already reserved. No booking or slots created for B.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-02 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-02

BK-03

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client B books 10:30, 30 min (overlaps the middle of A's)
Expected result	Rejected. This is the key overlap test; the original version of the app wrongly allowed it.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-03 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-03

BK-04

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client B books 09:00 for 60 min (ends exactly at 10:00)
Expected result	Accepted: back-to-back appointments are allowed.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-04 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-04

BK-05

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client B books 11:00
Expected result	Accepted (A's last block is 10:45 to 11:00).
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-05 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-05

BK-06

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client B books the same time at a different location
Expected result	Accepted: each location is its own calendar.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-06 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-06

BK-07

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A books another appointment overlapping her own
Expected result	Rejected with the 'overlaps your own booking' message.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-07 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-07

BK-08

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Two phones submit the same slot at the same moment (repeat three times)
Expected result	Only one succeeds (transaction).
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-08 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-08

BK-09

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Book on a closed day (for example Sunday)
Expected result	Rejected, message names the open days.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-09 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-09

BK-10

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Book on a South African public holiday (for example 25 December, 1 January), and a holiday in next year
Expected result	Rejected with the holiday message; the next-year date confirms the per-year fetch.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-10 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-10

BK-11

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Book before opening time; at 16:45 for 60 min when closing is 17:00
Expected result	Rejected with the correct reason (before opening; runs past closing).
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-11 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-11

BK-12

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Choose 16:00 with 30 min, then change duration to 60 min while closing is 17:00
Expected result	If it no longer fits, the time is cleared and the user told.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-12 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-12

BK-13

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Choose a time with minutes not on the quarter hour (10:07)
Expected result	Rejected.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-13 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-13

BK-14

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Book with no date selected; with no time
Expected result	Rejected with a message.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-14 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-14

BK-15

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Cancel Client A's booking (confirm dialog)
Expected result	Booking status becomes cancelled; all four slot documents are deleted in the same operation; reminder cancelled.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-15 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-15

BK-16

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client B immediately books the freed time
Expected result	Accepted.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-16 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-16

BK-17

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Cancel an already cancelled or completed booking
Expected result	Cancel button not offered, or a clear 'cannot be cancelled' message.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-17 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-17

BK-18

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Admin declines a pending booking (confirm dialog)
Expected result	Same result as BK-15 and the record is kept as cancelled.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-18 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-18

BK-19

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Admin deletes a booking (confirm dialog)
Expected result	Booking document is gone; slots are freed if it was still active; Client A's list updates.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-19 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-19

BK-20

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Delete an old cancelled booking whose time has since been re-booked by someone else
Expected result	The new booking's slots are not removed.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-20 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-20

BK-21

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Go offline, then submit a booking and cancel a booking
Expected result	Both blocked with the offline message; no partial data in Firestore.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-21 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-21

BK-22

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Book an Online appointment at 10:00, then another client books Online at 10:00
Expected result	Rejected, because all online bookings share one calendar. Record this as a design limitation (one online appointment at a time).
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR BK-22 HERE

App message plus the Firestore bookings / bookedSlots result (use two phones where needed)

Figure BK-22

6.5 Database permissions and access control — Ignatius Mayibongwe Khumalo

How to run

Run these two ways: (a) by using the app as the relevant user, and (b) in the Firebase Rules Playground, where you can simulate an authenticated request for a chosen uid. The Playground result is the proof for the report.

PM-01

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Unauthenticated read of bookings or users
Expected result	Denied.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-01 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-01

PM-02

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Unauthenticated read of practiceInfo
Expected result	Allowed (public by design, needed for the app before login).
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-02 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-02

PM-03

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A reads users/{A}
Expected result	Allowed.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-03 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-03

PM-04

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A reads users/{B}
Expected result	Denied.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-04 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-04

PM-05

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Admin reads any users document
Expected result	Allowed.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-05 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-05

PM-06

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Create users/{A} with role: "admin"
Expected result	Denied (create requires role client). This was a privilege-escalation hole in the first version.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-06 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-06

PM-07

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A updates own profile and changes role to admin
Expected result	Denied.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-07 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-07

PM-08

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A reads own booking
Expected result	Allowed.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-08 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-08

PM-09

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A reads Client B's booking
Expected result	Denied.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-09 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-09

PM-10

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A creates a booking with status confirmed, or paid true, or another user's clientId
Expected result	Denied.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-10 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-10

PM-11

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A updates own booking to status cancelled only
Expected result	Allowed.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-11 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-11

PM-12

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A updates own booking to confirmed, or sets paid: true, or changes the date
Expected result	Denied.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-12 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-12

PM-13

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A deletes a booking
Expected result	Denied.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-13 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-13

PM-14

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Admin updates status and paid; admin deletes a booking
Expected result	Allowed.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-14 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-14

PM-15

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A creates a slot with clientId = B's uid
Expected result	Denied.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-15 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-15

PM-16

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A deletes Client B's slot
Expected result	Denied.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-16 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-16

PM-17

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Anyone updates (modifies) a slot
Expected result	Denied.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-17 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-17

PM-18

Owner	Ignatius Mayibongwe Khumalo
Steps / what to check	Client A writes practiceInfo
Expected result	Denied. Admin write allowed.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR PM-18 HERE

Rules Playground result screenshot (simulated request and Allowed/Denied outcome)

Figure PM-18

6.6 Integration testing — All three members

How to run

Integration tests follow one complete story across the interface, authentication, Firestore and notifications. Run each with two phones (Client on one, Admin on the other) so the live updates can be seen.

INT-01

Owner	All three members
Steps / what to check	New client lifecycle
Expected result	Register, verify email, login, view services, book, see confirmation, see booking in My Bookings, Firestore documents correct.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR INT-01 HERE

Screenshots or a screen recording across both phones, plus the Firestore result

Figure INT-01

INT-02

Owner	All three members
Steps / what to check	Admin approval loop
Expected result	Client books; admin dashboard updates in real time; admin confirms; client's My Bookings changes to confirmed without a refresh; admin marks paid, then completed.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR INT-02 HERE

Screenshots or a screen recording across both phones, plus the Firestore result

Figure INT-02

INT-03

Owner	All three members
Steps / what to check	Cancellation loop
Expected result	Client cancels; admin dashboard shows cancelled; slots freed; second client can book the same time.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR INT-03 HERE

Screenshots or a screen recording across both phones, plus the Firestore result

Figure INT-03

INT-04

Owner	All three members
Steps / what to check	Admin changes practice hours
Expected result	Admin changes closing time; client booking screen enforces it on the next attempt.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR INT-04 HERE

Screenshots or a screen recording across both phones, plus the Firestore result

Figure INT-04

INT-05

Owner	All three members
Steps / what to check	Holiday API integration
Expected result	Booking blocked on a public holiday; then block the network and confirm the app warns that holidays could not be loaded and still enforces other rules.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR INT-05 HERE

Screenshots or a screen recording across both phones, plus the Firestore result

Figure INT-05

INT-06

Owner	All three members
Steps / what to check	Notifications
Expected result	Booking submitted shows an immediate notification; the reminder for the evening before is scheduled; turning reminders off in Settings cancels them; turning on reschedules; cancelling a booking removes its reminder.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR INT-06 HERE

Screenshots or a screen recording across both phones, plus the Firestore result

Figure INT-06

INT-07

Owner	All three members
Steps / what to check	Offline and recovery
Expected result	Go offline mid-use, then reconnect: banner appears and clears, nothing is duplicated, live lists catch up.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR INT-07 HERE

Screenshots or a screen recording across both phones, plus the Firestore result

Figure INT-07

INT-08

Owner	All three members
Steps / what to check	Session persistence
Expected result	Log in, kill the app, reopen offline: user is routed to the correct home screen using the cached role.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR INT-08 HERE

Screenshots or a screen recording across both phones, plus the Firestore result

Figure INT-08

INT-09

Owner	All three members
Steps / what to check	Analytics and export
Expected result	Admin Analytics numbers match what you count by hand in Firestore; CSV export contains the same bookings.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR INT-09 HERE

Screenshots or a screen recording across both phones, plus the Firestore result

Figure INT-09

INT-10

Owner	All three members
Steps / what to check	Two clients, one slot
Expected result	Clients A and B both try the final free slot at the same time: exactly one booking exists, both users see a correct result.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR INT-10 HERE

Screenshots or a screen recording across both phones, plus the Firestore result

Figure INT-10

INT-11

Owner	All three members
Steps / what to check	Search and filter
Expected result	Admin dashboard search and filters return the right bookings and update live when a new booking arrives.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR INT-11 HERE

Screenshots or a screen recording across both phones, plus the Firestore result

Figure INT-11

INT-12

Owner	All three members
Steps / what to check	Account changes
Expected result	Change email and password, then log out and in with the new details; bookings are still linked to the same user.
Result	**PENDING** mark after running: ☐ PASS ☐ FAIL
Device / date	☐ Phone ☐ Tablet Date: ____ / ____ / _____

INSERT SCREENSHOT FOR INT-12 HERE

Screenshots or a screen recording across both phones, plus the Firestore result

Figure INT-12

8. Defect log (B1 to B27)

B1 to B12 were found during development. B13 to B27 were found in the final implementation audit and fixed in the hardening phases.

#	Defect	Root cause	Resolution
B1	Admin Dashboard XML failed to compile after adding app:tint	Missing xmlns:app namespace on the root layout tag	Added the namespace declaration
B2	Logout unreachable on Settings	Buttons sat below the visible scroll area, no scroll indicator	Added a scroll thumb and repositioned logout
B3	Emulator 'Unknown error' on run	Emulator process had hung	Cold-booted the emulator
B4	PackageInfo import error	Wrong package path (android.pm vs android.content.pm)	Corrected the import
B5	Test files could not resolve JUnit/Espresso imports	Files placed in src/main/java instead of src/test / src/androidTest	Moved to the correct source sets
B6	strings.xml failed to parse	A stray invalid closing tag was accidentally included	Removed the invalid tag
B7	Admin login crashed after the date-filter redesign	tvFilterDate changed to a LinearLayout in XML but the Java field type was not updated	Changed the field type to View
B8	Register consent text clipped	Checkbox row height/text-wrap not configured	wrap_content rows, maxLines/ellipsize none, explicit bottom margins
B9	Status badges hard to read in light mode	Black text on dark badge backgrounds	Bold white badge text for all four statuses
B10	Back on Home returned to Welcome (looked like a logout)	Welcome stayed on the back stack; CLEAR_TOP did not clear it	Double-press-to-exit; navigation flags changed to NEW_TASK | CLEAR_TASK
B11	Register did not centre vertically on tablet	NestedScrollView + fillViewport + weight-based centring does not reliably stretch	Plain ScrollView with a constrained content column
B12	Register header dragged to the middle on tablet	Header and form shared one centred container	Layout restructured so the header stays at the top
B13	A user could register themselves as an admin	users create rule checked ownership but not the role value	Rule now requires role == client on create
B14	A client could confirm their own booking or mark it paid	bookings update rule allowed the owner to change any field	Clients limited to status to cancelled; all else admin-only
B15	Any user could overwrite or delete any time slot	bookedSlots allowed any signed-in user to write	Slots carry clientId; create/delete limited to owner or admin; update denied
B16	Cancel/decline could leave a stuck slot	Status update and slot delete were two separate calls	One Firestore transaction (BookingActions)
B17	Bookings had no true Delete	Cancel only set status to cancelled	Admin Delete Booking (permanent, confirmed, atomic)
B18	Two different clients could book overlapping times	Overlap only checked the same client's bookings; slot IDs ignored duration	Per-15-minute block reservation inside a transaction
B19	An appointment could finish after closing time	Only the start hour was compared with closing time	Start + duration checked; re-checked on duration change and submit
B20	Admin's Operating Days setting had no effect	Booking screen hardcoded Saturday/Sunday	Structured openDays list with checkboxes, enforced when booking
B21	Booking Reminders switch did nothing	Preference saved but never read before notifying	Enforced in NotificationHelper; reminders cancelled/rescheduled on toggle

#	Defect	Root cause	Resolution
B22	Failed registration could leave an orphaned login	Auth account created, then profile write failed with no rollback	Auth account deleted if the profile write fails
B23	Booking with no status could crash screens	Unguarded switch on a null status	StatusUtils treats missing status as pending
B24	Holidays only checked for the current year	Holidays fetched once, for the current year	Fetched and cached per year
B25	Email validation accepted '@' and 'a@b'	Rule was contains('@')	Regex for local part, @ and dotted domain; tests updated
B26	Default test stubs and .idea/ committed	Never removed	Deleted; .idea untracked and ignored
B27	Opening Settings could show a false 'reminders enabled' toast	Switch listener attached before the saved value loaded	Listener attached only after the saved value is applied

9. Sign-off

We confirm that the tests above were run on the devices listed on the first page and that the screenshots are genuine results from those runs.

Student Signature(s)	Date
Ignatius Mayibongwe Khumalo	
Moqechane, Khauhelo	
Faith Murau	